101 Greenwood Avenue, Ste 170
Jenkintown, PA 19046
+1 (855) 572-8863

Pioneer Research Institute Evaluation Form

Scholar Information

Scholar's Full Name:	Panayappan, Ananya Arun
Scholar's Research Concentration:	The Random Walk of Mathematical Finance
Title of Scholar's Research Paper:	A Quantitative Framework for Portfolio Resilience: Insights from Indian Markets during COVID-19 and Demonetization

Grading and Academic Oversight Section

Grade conferred by professor: A

**Full grading rubric approved by Pioneer Academics and Oberlin College*

The Pioneer Research Institute's college accreditation and academic oversight are conducted collaboratively with Oberlin College:

Listing on Common Application:	"Summer Program, Credit Awarded Directly by Oberlin College"
Course Title on Coalition Application:	"099" "Pioneer Research Course"
College Course ID on UC Application:	"INST 099" "Pioneer Research Course"

Comprehensive Evaluation Section

Evaluator Name & Full Title:	Hesam Oveys, Clinical Assistant Professor
College/University & Department:	New York University, Courant Institute Dept. of Mathematics
Full Educational Background of Evaluator:	Ph.D., Mathematics, University of Missouri (2015)

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Please briefly explain the nature and requirements of the research paper and your interaction with the scholar:

The research in this institute focused on the core principles of mathematical finance. In the initial group sessions, scholars learned the foundational concepts of probability theory, stochastic processes, and financial instruments. We covered key models for asset pricing and computational techniques like Monte Carlo simulations. The culminating project was a formal research paper of at least 15 pages where each scholar had to formulate a unique research question, apply the quantitative methods learned, and present their findings in a clear, well-structured academic format.
My interaction with Ananya was exceptionally professional. During group sessions, she demonstrated a quick and thorough grasp of the core concepts. When we began our individual sessions, she arrived with a clear, ambitious, and well-defined research goal: to create a novel quantitative index for portfolio resilience. Ananya consistently drove the project forward with a high degree of autonomy, bringing fully-formed analysis to our meetings for discussion. My role was primarily to act as a consultant, offering feedback on a project that she managed with the skill of a seasoned researcher.

Please rate the scholar in the following areas:

	Excellent	Good	Average	Below Average	Poor
Ability to form original ideas and concepts	**X**				
Ability to communicate thoughts in an effective and articulate manner	**X**				
Ability to synthesize and organize information from disparate sources	**X**				
Level of scholar's curiosity, aptitude, and industriousness	**X**				

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Based on the scholar's performance throughout Pioneer, how would you rate this scholar's potential for undergraduate-level academic work at a top college/university, relative to other undergraduate students whose academic work you have mentored?

One of the best I have encountered, top 5% **X**	Excellent, top 10%
Very Good, top 25%	Good, top 50%
Below Average	

Describe some of the scholar's strengths:

Ananya's greatest strength is her ability to design and execute a complete, end-to-end quantitative research project. She successfully developed a new composite metric—the "Resilience Index," as she calls it—which required thoughtfully selecting, weighting, and normalizing multiple financial indicators. Second, her analytical approach is rigorous and practical; she didn't just propose the index but immediately applied it to portfolio optimization, demonstrating its tangible value by showing it improved risk-adjusted returns. Finally, Ananya exhibits exceptional scientific foresight by generalizing her index and validating it on out-of-sample events, which is a critical step for proving a model's robustness.

Describe some areas in which the scholar can improve:

A key area for improvement lies in justifying the specific construction of the Resilience Index more formally. The weights for the index components were chosen based on "heuristics and economic judgement," which is a valid starting point. To make this publishable in a journal, a more rigorous justification is probably going to be needed, such as conducting a principal component analysis (PCA) to derive the weights empirically or performing a more extensive sensitivity analysis to show that the results are robust to different weighting schemes. However, that's well beyond what was expected, but this would make the construction of the index less subjective and more defensible.

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While the second half of the research work was one-to-one with the professor, the first half involved a small research cohort of 3-6 scholars. Describe the contribution that the scholar made to the research group discussion and learning:

The initial group sessions were primarily lecture-based to cover the required mathematical background. In this setting, Ananya was a model participant—attentive, prepared, and engaged. Her questions were precise and demonstrated a deep level of comprehension. Her consistent high performance on assignments and her clear focus set a benchmark for academic excellence within the cohort.

Would you recommend this scholar to a college admissions officer at a top college or university? Why or why not?

Yes, I recommend Ananya with my strongest possible endorsement. Her research paper is not simply a student project; it is work that approaches the level of an undergraduate honors thesis in quantitative finance. She has demonstrated a good grasp of data analysis, portfolio theory, and the scientific method. Her ability to conceptualize a new financial metric, implement it computationally, and validate its utility is impressive. Ananya will undoubtedly excel and become a positive addition at any elite university.

Evaluator's Signature:	
Evaluator's Printed Name:	Hesam Oveys
Date of Evaluation:	9/4/2025
Evaluator's Email:	hesamo@gmail.com

Pioneer Academics affirms that the scholar whose research paper is attached has followed Pioneer Academics' protocols for developing original research and that their paper has met Pioneer Academics' standards for authenticity. The grading and evaluation have met the standards collaboratively defined by Pioneer Academics and Oberlin College & Conservatory. If there are any questions regarding the work's academic integrity, please contact us at academic.integrity@pioneeracademics.com or by calling (855) 572-8863. We will conduct an immediate review and respond to your questions promptly.